

Tutorial 6 — Polynomial Interpolation

Solutions

Numerical Methods and Optimization

ENSIA — 3rd Year

Spring 2026

Exercise 1 — Statement

We are given the interpolation data

$$x = [-2, 0, 1, 2], \quad y = [4, 0, 0, 4].$$

We want to determine which of the following is the interpolation polynomial:

$$P_1(X) = X^4 - \frac{2}{3}X^3 - 3X^2 + \frac{8}{3}X$$

$$P_2(X) = \frac{3}{4}X^2 - \frac{4}{3}$$

$$P_3(X) = \frac{1}{3}X^3 + X^2 - \frac{4}{3}X$$

Exercise 1 — Checking the candidates

The interpolation polynomial P must satisfy

$$P(-2) = 4, \quad P(0) = 0, \quad P(1) = 0, \quad P(2) = 4.$$

1) Check P_2 :

$$P_2(0) = \frac{3}{4} \cdot 0^2 - \frac{4}{3} = -\frac{4}{3} \neq 0.$$

So P_2 is not correct.

2) Check P_3 :

$$P_3(0) = 0,$$

$$P_3(1) = \frac{1}{3} + 1 - \frac{4}{3} = 0,$$

$$P_3(2) = \frac{8}{3} + 4 - \frac{8}{3} = 4,$$

$$P_3(-2) = -\frac{8}{3} + 4 + \frac{8}{3} = 4.$$

Hence P_3 passes through all four points.

Exercise 1 — Why P_3 is the interpolation polynomial

3) Check P_1 :

$$P_1(0) = 0, \quad P_1(1) = 0, \quad P_1(2) = 4, \quad P_1(-2) = 4.$$

So P_1 also passes through the four given points.

However, for 4 distinct nodes, the interpolation polynomial is the unique polynomial of degree at most 3.

- P_3 has degree 3,
- P_1 has degree 4.

Therefore the interpolation polynomial is

$$P(X) = P_3(X) = \frac{1}{3}X^3 + X^2 - \frac{4}{3}X.$$

Exercise 1 — Extra remark

Why does P_1 also interpolate the same data?

$$\begin{aligned}P_1(X) - P_3(X) &= \left(X^4 - \frac{2}{3}X^3 - 3X^2 + \frac{8}{3}X\right) - \left(\frac{1}{3}X^3 + X^2 - \frac{4}{3}X\right) \\&= X^4 - X^3 - 4X^2 + 4X = X(X^3 - X^2 - 4X + 4) \\&= X(X - 1)(X^2 - 4) = X(X - 1)(X - 2)(X + 2).\end{aligned}$$

Thus

$$P_1(X) = P_3(X) + X(X - 1)(X - 2)(X + 2).$$

The added term vanishes at

$$X = -2, 0, 1, 2,$$

so P_1 and P_3 have the same values at the interpolation nodes.

P_1 and P_3 coincide on the nodes, but only P_3 has degree ≤ 3 .

Exercise 2 — Statement

- 1) Show that there are infinitely many polynomials of degree 2 whose graph passes through the points $(0, 0)$ and $(1, 0)$.
- 2) Show that there does not exist a polynomial of degree 2 passing through the points $(0, 1)$, $(1, 4)$, $(2, 15)$, $(3, 40)$.

Exercise 2.1 — Infinitely many quadratic polynomials

Let

$$P(x) = ax^2 + bx + c, \quad a \neq 0,$$

since we want a polynomial of degree 2.

Because the graph passes through $(0,0)$,

$$P(0) = c = 0.$$

So

$$P(x) = ax^2 + bx.$$

Because the graph also passes through $(1,0)$,

$$P(1) = a + b = 0,$$

hence

$$b = -a.$$

Therefore

$$P(x) = ax^2 - ax = ax(x - 1).$$

Exercise 2.1 — Conclusion

For every real number $a \neq 0$,

$$P(x) = ax(x-1)$$

is a polynomial of degree 2, and

$$P(0) = 0, \quad P(1) = 0.$$

So it passes through the points $(0, 0)$ and $(1, 0)$.

Since there are infinitely many choices of $a \in \mathbb{R} \setminus \{0\}$, there are infinitely many such quadratic polynomials.

$$P(x) = ax(x-1), \quad a \in \mathbb{R} \setminus \{0\}$$

Exercise 2.2 — Assume a quadratic polynomial exists

Suppose there exists a polynomial of degree 2

$$P(x) = ax^2 + bx + c$$

such that, $a \neq 0$ and

$$P(0) = 1, \quad P(1) = 4, \quad P(2) = 15, \quad P(3) = 40.$$

Using the first three conditions:

$$P(0) = c = 1$$

Then

$$P(1) = a + b + 1 = 4 \implies a + b = 3.$$

And

$$P(2) = 4a + 2b + 1 = 15 \implies 4a + 2b = 14 \implies 2a + b = 7.$$

Exercise 2.2 — Contradiction

Solving the system

$$a + b = 3, \quad 2a + b = 7.$$

we get

$$a = 4.$$

and

$$b = 3 - a = 3 - 4 = -1.$$

So the only possible quadratic polynomial is

$$P(x) = 4x^2 - x + 1.$$

Now evaluate it at $x = 3$:

$$P(3) = 4 \cdot 9 - 3 + 1 = 36 - 3 + 1 = 34.$$

But the required value is

$$P(3) = 40.$$

This is impossible. Therefore no polynomial of degree 2 passes through the four given points.

Exercise 3 — Statement

Compute the Lagrange interpolation polynomials for the following data:

(a)

$$x = [-1, 2, 3], \quad y = [4, 4, 8]$$

(b)

$$x = [-2, -1, 0, 1], \quad y = [0, -2, -4, 0]$$

(c)

$$x = [-1, 0, 1, 2], \quad y = [6, 2, 0, 0]$$

Exercise 3(a)

The interpolation points are

$$(-1, 4), \quad (2, 4), \quad (3, 8).$$

Since

$$P(-1) = 4 \quad \text{and} \quad P(2) = 4,$$

we know that $P(x) - 4$ is a second degree polynomial and has roots -1 and 2 .

So we write

$$P(x) = 4 + a(x + 1)(x - 2).$$

Now use the third condition:

$$P(3) = 8.$$

Thus

$$8 = 4 + a(3 + 1)(3 - 2) = 4 + 4a \implies a = 1$$

Therefore

$$\boxed{P(x) = x^2 - x + 2.}$$

Exercise 3(b)

The interpolation points are

$$(-2, 0), \quad (-1, -2), \quad (0, -4), \quad (1, 0).$$

Since

$$P(-2) = 0 \quad \text{and} \quad P(1) = 0,$$

the polynomial has factors $(x + 2)$ and $(x - 1)$.

Because the interpolation polynomial has degree at most 3, we write

$$P(x) = (x + 2)(x - 1)(ax + b).$$

Exercise 3(b)

Now we use the remaining values.

$$P(0) = 2(-1)b = -2b = -4 \implies b = 2.$$

$$P(-1) = 1(-2)(-a + b) = -2(-a + b) = -2. \implies a = 1.$$

Hence

$$P(x) = (x + 2)^2(x - 1) = x^3 + 3x^2 - 4.$$

Exercise 3(c)

The interpolation points are

$$(-1, 6), \quad (0, 2), \quad (1, 0), \quad (2, 0).$$

Since

$$P(1) = 0 \quad \text{and} \quad P(2) = 0,$$

the polynomial is at most of degree 3 and has factors $(x - 1)$ and $(x - 2)$.

So we write

$$P(x) = (x - 1)(x - 2)(ax + b).$$

Exercise 3(c)

Now use the other two conditions.

$$P(0) = (-1)(-2)b = 2b = 2 \implies b = 1.$$

$$P(-1) = (-2)(-3)(-a + b) = 6(-a + b) = 6. \implies a = 0.$$

Therefore

$$P(x) = (x - 1)(x - 2) = x^2 - 3x + 2.$$

Exercise 4 — Statement

Let P be a polynomial, and let $x_0, \dots, x_n \in \mathbb{R}$ be interpolation nodes. Define

$$\pi_n(x) = (x - x_0)(x - x_1) \cdots (x - x_n).$$

Show that the interpolation polynomial of P at the nodes

$$x_0, \dots, x_n$$

is the remainder of the Euclidean division of P by $\pi_n(x)$.

Exercise 4 — Euclidean division

Using euclidean division we know that there exist two polynomials Q and R such that

$$P(x) = Q(x)\pi_n(x) + R(x),$$

with

$$\deg(R) < \deg(\pi_n) = n + 1.$$

Therefore

$$\deg(R) \leq n.$$

So R is a polynomial of degree at most n .

Exercise 4 — Values at the interpolation nodes

For every node x_i , we have

$$\pi_n(x_i) = 0,$$

and consequently

$$P(x_i) = R(x_i) \quad \text{for all } i = 0, \dots, n.$$

So R and P have the same values at all interpolation nodes.

We conclude that R is the interpolation polynomial.

Exercise 5 — Statement

Let $x_0 = 1$, $x_1 = 2$, $x_2 = 3$, $x_3 = 4$, $x_4 = 5$, $x_5 = 6$.

- 1) Recall the expression of the Newton basis of $\mathbb{R}_5[X]$ associated with the nodes 1, 2, 3, 4, 5, 6.
- 2) Show that it is indeed a basis.
- 3) Give, in this basis, the expression of the polynomial $P \in \mathbb{R}_5[X]$ such that

$$P(1) = P(6) = 1, \quad P(2) = P(3) = P(4) = P(5) = 0.$$

Exercise 5.1 — Newton basis

The Newton basis associated with the nodes

$$1, 2, 3, 4, 5, 6$$

is defined by

$$N_0(x) = 1,$$

$$N_1(x) = x - 1,$$

$$N_2(x) = (x - 1)(x - 2),$$

$$N_3(x) = (x - 1)(x - 2)(x - 3),$$

$$N_4(x) = (x - 1)(x - 2)(x - 3)(x - 4),$$

$$N_5(x) = (x - 1)(x - 2)(x - 3)(x - 4)(x - 5).$$

So the Newton basis is

$$(N_0, N_1, N_2, N_3, N_4, N_5).$$

Exercise 5.2 — Proof that it is a basis

We prove that $(N_0, \dots, N_5)$ is linearly independent.

Assume

$$a_0 N_0(x) + a_1 N_1(x) + a_2 N_2(x) + a_3 N_3(x) + a_4 N_4(x) + a_5 N_5(x) = 0 \quad \forall x \in \mathbb{R}.$$

Evaluate successively at the nodes:

At $x = 1$,

$$N_0(1) = 1, \quad N_k(1) = 0 \text{ for } k \geq 1,$$

hence

$$a_0 = 0.$$

At $x = 2$,

$$a_1 N_1(2) = 0, \quad N_1(2) = 1,$$

so

$$a_1 = 0.$$

Exercise 5.2 — Continuation of the proof

Continuing in the same way:

At $x = 3$,

$$a_2 N_2(3) = 0, \quad N_2(3) = (3 - 1)(3 - 2) = 2 \neq 0,$$

so

$$a_2 = 0.$$

At $x = 4$, we get

$$a_3 = 0.$$

At $x = 5$, we get

$$a_4 = 0.$$

At $x = 6$, we get

$$a_5 = 0.$$

Therefore

$$a_0 = a_1 = a_2 = a_3 = a_4 = a_5 = 0.$$

So $(N_0, \dots, N_5)$ is linearly independent.

Since $\dim(\mathbb{R}_5[X]) = 6$, this family of 6 polynomials is a basis of $\mathbb{R}_5[X]$.

$(N_0, \dots, N_5)$ is a basis of $\mathbb{R}_5[X]$.

Exercise 5.3 — Divided differences table

We interpolate the data

$$(1, 1), (2, 0), (3, 0), (4, 0), (5, 0), (6, 1).$$

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$	$f[x_i, \dots, x_{i+4}]$	$f[x_i, \dots, x_{i+5}]$
1	1	-1	$\frac{1}{2}$	$-\frac{1}{6}$	$\frac{1}{24}$	0
2	0	0	0	0	$\frac{1}{24}$	
3	0	0	0	$\frac{1}{6}$		
4	0	0	$\frac{1}{2}$			
5	0	1				
6	1					

Exercise 5.3 — Newton coefficients

The Newton coefficients are the first entries of each column:

$$f[1] = 1, \quad f[1, 2] = -1, \quad f[1, 2, 3] = \frac{1}{2},$$

$$f[1, 2, 3, 4] = -\frac{1}{6}, \quad f[1, 2, 3, 4, 5] = \frac{1}{24}, \quad f[1, 2, 3, 4, 5, 6] = 0.$$

Hence, in the Newton basis,

$$\begin{aligned} P(x) = & 1 - (x - 1) + \frac{1}{2}(x - 1)(x - 2) \\ & - \frac{1}{6}(x - 1)(x - 2)(x - 3) + \frac{1}{24}(x - 1)(x - 2)(x - 3)(x - 4). \end{aligned}$$

$$P(x) = 1 - (x - 1) + \frac{1}{2}(x - 1)(x - 2) - \frac{1}{6}(x - 1)(x - 2)(x - 3) + \frac{1}{24}(x - 1)(x - 2)(x - 3)(x - 4)$$

Exercise 6 — Statement

Compute the Lagrange interpolation polynomial of

$$f(x) = x(x^2 - 1) = x^3 - x$$

with respect to the points

$$x_0 = -1, \quad x_1 = 1, \quad x_2 = 2.$$

Same question, adding the point

$$x_3 = -2.$$

Exercise 6.1 — Method 1: Lagrange formula

We have

$$f(-1) = 0, \quad f(1) = 0, \quad f(2) = 6.$$

The Lagrange polynomials are

$$L_0(x) = \frac{(x-1)(x-2)}{(-1-1)(-1-2)} = \frac{(x-1)(x-2)}{6},$$

$$L_1(x) = \frac{(x+1)(x-2)}{(1+1)(1-2)} = -\frac{(x+1)(x-2)}{2},$$

$$L_2(x) = \frac{(x+1)(x-1)}{(2+1)(2-1)} = \frac{(x+1)(x-1)}{3}.$$

So

$$P_2(x) = f(-1)L_0(x) + f(1)L_1(x) + f(2)L_2(x).$$

Hence

$$P_2(x) = 2x^2 - 2.$$

Exercise 6.1 — Method 2: Using Exercise 4

We use

$$\pi_2(x) = (x + 1)(x - 1)(x - 2).$$

Since

$$f(x) = x(x^2 - 1) = x(x - 1)(x + 1),$$

Then

$$\begin{aligned} f(x) &= ((x - 2) + 2)(x - 1)(x + 1) \\ &= (x - 2)(x - 1)(x + 1) + 2(x - 1)(x + 1). \end{aligned}$$

So

$$f(x) = \pi_2(x) + 2(x^2 - 1).$$

Therefore, the remainder of the Euclidean division of f by π_2 is

$$2(x^2 - 1).$$

By Exercise 4,

$$P_2(x) = 2x^2 - 2.$$

Exercise 6.2 — Adding the point -2

Now we interpolate at the four points

$$-2, \quad -1, \quad 1, \quad 2.$$

Since

$$f(x) = x^3 - x$$

is itself a polynomial of degree 3, and since the interpolation polynomial through 4 distinct points is unique in $\mathbb{R}_3[X]$, it follows that f itself is the interpolation polynomial.

Therefore

$$P_3(x) = x^3 - x = x(x^2 - 1).$$

Exercise 7 — Statement

Let

$$f(x) = \cos x, \quad x \in [0, 1].$$

Estimate the minimum number of interpolation nodes such that the error between f and its Lagrange interpolation polynomial is less than

0.01.

Interpolation error formula

Let P_n be the interpolation polynomial of f at the distinct nodes

$$x_0, \dots, x_n,$$

and let

$$\omega_{n+1}(t) = \prod_{i=0}^n (t - x_i).$$

Then for every $x \neq x_0, \dots, x_n$, there exists ξ such that

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x).$$

Hence

$$|f(x) - P_n(x)| \leq \frac{\max_{t \in [a,b]} |f^{(n+1)}(t)|}{(n+1)!} |\omega_{n+1}(x)|.$$

Quick proof using Rolle's theorem

Fix $x \notin \{x_0, \dots, x_n\}$, and define

$$\lambda = \frac{f(x) - P_n(x)}{\omega_{n+1}(x)}.$$

Consider

$$\varphi(t) = f(t) - P_n(t) - \lambda \omega_{n+1}(t).$$

Then

$$\varphi(x_i) = 0 \quad (i = 0, \dots, n), \quad \varphi(x) = 0.$$

So φ has $n + 2$ zeros. By applying Rolle's theorem $n + 1$ times, there exists ξ such that

$$\varphi^{(n+1)}(\xi) = 0.$$

But

$$\varphi^{(n+1)}(t) = f^{(n+1)}(t) - \lambda(n+1)!,$$

which gives

$$f(x) - P_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} \omega_{n+1}(x).$$

Exercise 7 — Error estimate

Here

$$f(x) = \cos x.$$

So

$$|f^{(n+1)}(t)| \leq 1 \quad \text{for all } t \in [0, 1].$$

Also, since $x, x_i \in [0, 1]$,

$$|x - x_i| \leq 1,$$

hence

$$|\omega_{n+1}(x)| = \prod_{i=0}^n |x - x_i| \leq 1.$$

Therefore

$$|f(x) - P_n(x)| \leq \frac{1}{(n+1)!}.$$

We want

$$\frac{1}{(n+1)!} < 0.01 = \frac{1}{100},$$

Exercise 7 — Conclusion

Now

$$4! = 24 < 100, \quad 5! = 120 > 100.$$

So the smallest possible value is

$$n + 1 = 5.$$

Therefore the minimum number of interpolation nodes given by this estimate is

5.

So we need at least 5 interpolation nodes to guarantee

$$|f(x) - P_n(x)| < 0.01.$$

Exercise 8 — Statement

A social media platform collected the following data:

x	1	2	4	5
y	0.12	0.20	0.35	0.42

- 1 Construct the interpolation polynomial using the Lagrange method.
- 2 Construct the interpolation polynomial using the Newton divided differences method.
- 3 Estimate the click probability at $x = 3$ and $x = 4.5$.
- 4 Compare with linear interpolation between nearest points.
- 5 Explain why interpolation is useful when training data are limited.

Exercise 8.1 — Lagrange interpolation

The interpolation points are

$$(1, 0.12), \quad (2, 0.20), \quad (4, 0.35), \quad (5, 0.42).$$

The Lagrange polynomial is

$$P(x) = \sum_{i=0}^3 y_i L_i(x),$$

with

$$L_0(x) = \frac{(x-2)(x-4)(x-5)}{(1-2)(1-4)(1-5)} = -\frac{(x-2)(x-4)(x-5)}{12},$$

$$L_1(x) = \frac{(x-1)(x-4)(x-5)}{(2-1)(2-4)(2-5)} = \frac{(x-1)(x-4)(x-5)}{6},$$

$$L_2(x) = \frac{(x-1)(x-2)(x-5)}{(4-1)(4-2)(4-5)} = -\frac{(x-1)(x-2)(x-5)}{6},$$

$$L_3(x) = \frac{(x-1)(x-2)(x-4)}{(5-1)(5-2)(5-4)} = \frac{(x-1)(x-2)(x-4)}{12}.$$

Exercise 8.1 — Simplified polynomial

So

$$P(x) = 0.12L_0(x) + 0.20L_1(x) + 0.35L_2(x) + 0.42L_3(x).$$

That is,

$$P(x) = -0.01(x-2)(x-4)(x-5) + \frac{1}{30}(x-1)(x-4)(x-5) \\ - \frac{7}{120}(x-1)(x-2)(x-5) + \frac{7}{200}(x-1)(x-2)(x-4).$$

After simplification,

$$P(x) = \frac{-x^2 + 51x + 22}{600}$$

or equivalently

$$P(x) = -\frac{x^2}{600} + \frac{17}{200}x + \frac{11}{300}.$$

Exercise 8.2 — Newton divided differences

We compute the divided differences:

x_i	$f[x_i]$	$f[x_i, x_{i+1}]$	$f[x_i, x_{i+1}, x_{i+2}]$	$f[x_i, \dots, x_{i+3}]$
1	0.12	0.08	$-\frac{1}{600}$	0
2	0.20	0.075	$-\frac{1}{600}$	
4	0.35	0.07		
5	0.42			

Therefore

$$P(x) = 0.12 + 0.08(x - 1) - \frac{1}{600}(x - 1)(x - 2).$$

Hence

$$P(x) = 0.12 + 0.08(x - 1) - \frac{1}{600}(x - 1)(x - 2).$$

Exercise 8.3 — Estimated values

Using

$$P(x) = \frac{-x^2 + 51x + 22}{600},$$

we get

$$P(3) = \frac{-9 + 153 + 22}{600} = \frac{166}{600} = \frac{83}{300} \approx 0.2767.$$

So

$$P(3) \approx 0.2767.$$

Also,

$$P(4.5) = \frac{-(4.5)^2 + 51(4.5) + 22}{600} = \frac{231.25}{600} \approx 0.3854.$$

So

$$P(4.5) \approx 0.3854.$$

Exercise 8.4 — Comparison with linear interpolation

For $x = 3$, we interpolate linearly between $x = 2$ and $x = 4$:

$$P_{\text{lin}}(3) = 0.20 + \frac{3-2}{4-2}(0.35 - 0.20) = 0.20 + \frac{1}{2}(0.15) = 0.275.$$

So

$$P_{\text{lin}}(3) = 0.275.$$

For $x = 4.5$, we interpolate linearly between $x = 4$ and $x = 5$:

$$P_{\text{lin}}(4.5) = 0.35 + \frac{4.5-4}{5-4}(0.42 - 0.35) = 0.35 + 0.5(0.07) = 0.385.$$

So

$$P_{\text{lin}}(4.5) = 0.385.$$

Exercise 8.5 — Interpretation

Interpolation helps when the available data are limited because it allows the system to estimate values at points that were not directly observed.

Here, the platform only knows the click probabilities for

$$x = 1, 2, 4, 5,$$

but it can still estimate the probability for

$$x = 3 \quad \text{or} \quad x = 4.5.$$

So interpolation provides a simple way to:

- fill the gaps between known data,
- obtain smooth predictions,
- help decision-making even with a small dataset.

Exercise 9 — Statement

Let

$$f(x) = \sin x,$$

and let $P_2(x)$ be the interpolation polynomial of f at the points

$$x_0 = 0, \quad x_1 = \frac{\pi}{4}, \quad x_2 = \frac{\pi}{2}.$$

Bound the error

$$\left| f\left(\frac{\pi}{5}\right) - P_2\left(\frac{\pi}{5}\right) \right|.$$

Exercise 9 — Error formula

For interpolation at three nodes, the error is

$$f(x) - P_2(x) = \frac{f^{(3)}(\xi)}{3!}(x - x_0)(x - x_1)(x - x_2)$$

for some ξ in the interval.

Here

$$f(x) = \sin x, \quad f^{(3)}(x) = -\cos x.$$

So at $x = \frac{\pi}{5}$,

$$f\left(\frac{\pi}{5}\right) - P_2\left(\frac{\pi}{5}\right) = \frac{-\cos(\xi)}{6} \left(\frac{\pi}{5} - 0\right) \left(\frac{\pi}{5} - \frac{\pi}{4}\right) \left(\frac{\pi}{5} - \frac{\pi}{2}\right).$$

Exercise 9 — Bound of the error

Since

$$|\cos(\xi)| \leq 1,$$

we obtain

$$\left| f\left(\frac{\pi}{5}\right) - P_2\left(\frac{\pi}{5}\right) \right| \leq \frac{1}{6} \left| \frac{\pi}{5} \left(\frac{\pi}{5} - \frac{\pi}{4} \right) \left(\frac{\pi}{5} - \frac{\pi}{2} \right) \right|.$$

Now

$$\frac{\pi}{5} - \frac{\pi}{4} = -\frac{\pi}{20}, \quad \frac{\pi}{5} - \frac{\pi}{2} = -\frac{3\pi}{10}.$$

Hence

$$\left| \frac{\pi}{5} \left(-\frac{\pi}{20} \right) \left(-\frac{3\pi}{10} \right) \right| = \frac{3\pi^3}{1000}.$$

Therefore

$$\left| f\left(\frac{\pi}{5}\right) - P_2\left(\frac{\pi}{5}\right) \right| \leq \frac{1}{6} \cdot \frac{3\pi^3}{1000} = \frac{\pi^3}{2000}.$$

Thus

$$\left| \sin\left(\frac{\pi}{5}\right) - P_2\left(\frac{\pi}{5}\right) \right| \leq \frac{\pi^3}{2000}$$